

Mathematical modelling and simulation for planning critical care capacity★

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Summary

Using average number of patients expected in a year, average length of stay and a target occupancy level to calculate the number of critical care beds needed is mathematically incorrect because of nonlinearity and variability in the factors that control length of stay. For a target occupancy in excess of 80%, this simple calculation will typically underestimate the number of beds required. More seriously, it provides no quantitative guidance information about other aspects of critical care demand such as the numbers of emergency patients transferred, deferral rates for elective patients and overall utilisation. The combination of appropriately analysing raw data and detailed mathematical modelling provides a much better method for estimating numbers of beds required. We describe this modelling approach together with evidence of its performance.

Keywords *Intensive care: organisation and administration, trends, utilisation.*

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The organisation and capacities of critical care units are important national concerns. Inadequate capacity in a critical care unit leads to cancellation of planned surgery, transfers of emergency patients and premature discharge. These are highly undesirable events which degrade the quality of care delivered by a hospital and may jeopardise outcome. However, because critical care is very expensive, excess capacity should be minimised so that wasteful under-utilisation is avoided.

The demand for critical care stems from many areas, hence the heterogeneity seen in nearly all aspects of critical care delivery. With regard to capacity planning, two of the most important parameters are the patients' arrival rates and patterns, and their length of stay on the critical care unit. The arrival rates vary from day to day and within a day; however, for most general critical care units, most patients arrive 'unpredictably' as emergencies. Patients' length of stay varies considerably and the

distribution is highly skewed with most patients staying only a few days and occasional patients having very long admissions. It is a well-known mathematical fact that required capacities in the face of non-linearity and considerable variability depend not only on the average values but also the amount of variability [1]. Organisations that only use average values to calculate required capacity will obtain misleading results, and if target occupancy exceeds about 80% in these simple calculations, then required capacity will be underestimated, as indicated in Appendix.

Critical care involves uncertainty, variability, complexity, numerous constraints and expensive resources, and there are many options for its organisation and management. It is very difficult, if at all possible, to use randomised controlled trials to obtain the necessary evidence to inform decisions about effective and efficient provision of critical care and this includes the optimum

number of beds. Metcalfe & Mcpherson [2] reported that the provision of intensive care beds varied considerably between hospitals and so there may be hospitals with under- and over-provision of critical care. Sufficiently detailed mathematical models can provide quantitative information for evaluating the practical range of options for critical care capacity and thus help decision-making at policy and operational levels. Models for studying allocation of hospital beds and capacity planning outside critical care have been considered by a number of researchers; examples are Kao & Tung [3], Smith-Daniels *et al.* [4], Parry & Petroda [5] and Harper & Shahani [6]. Williams [7] and Ridge *et al.* [8] have explored mathematical modelling for critical care, but have not provided evidence of the models' validity. Here, we extend these models for use in critical care. We discuss detailed mathematical models that can inform decision-making about the number of beds required for a critical care unit to meet its own individual workload. Case studies illustrate the necessary data analysis and the information provided by the models.

Methods

The models are set at the individual patient level and can be tuned to mimic the flow of patients through a particular critical care unit. The models require information about patients' categories, arrival patterns and lengths of stay. The use of the model may be described in four phases. First, some simple rules are required to describe the flow of patients through the critical care unit. Second, statistical information about current casemix, arrival patterns, lengths of stay and number of beds is needed. Third, the model is run repetitively, with the current rules dictating patients' flow, to simulate the working of the unit with patients arriving and leaving over long

periods. Fourth, the statistical information produced by the models can be compared with the unit's actual data to check that the model does reflect the particular critical care unit and its casemix.

Flow of patients through the critical care unit

The model requires a set of simple rules to reflect the flow of patients through the critical care area. We attempted to keep the rules as simple as possible and to reflect, as far as possible, what might be considered usual practice for a typical general adult critical care service. Admission criteria are applied to each of the presenting patients and, capacity allowing, most patients are admitted. If admission is not possible, then emergency patients will, typically, be sent to another hospital or nursed temporarily elsewhere in the same hospital, for example, the operating theatres' recovery area (Fig. 1). Planned admissions of patients will usually be deferred. Many hospitals have a policy of giving emergency patients a higher priority than planned cases. Some hospitals change the status of a planned admission to have the same priority as an emergency one if the affected patient has been deferred for a defined number of times. These factors can all be programmed into the model. In the present model, planned cases were given the same priority as emergency ones if their surgery had been cancelled three times.

Data required

The data required depend on the scenarios that have to be evaluated but for capacity planning, they fall into three groups. The first concerns patients' demographic details that are likely to affect length of stay. In our validation exercise we assumed that admission status (planned, emergency or previously deferred), patient source (operating theatres, accident and emergency department, general wards, other hospital or specialised area), severity

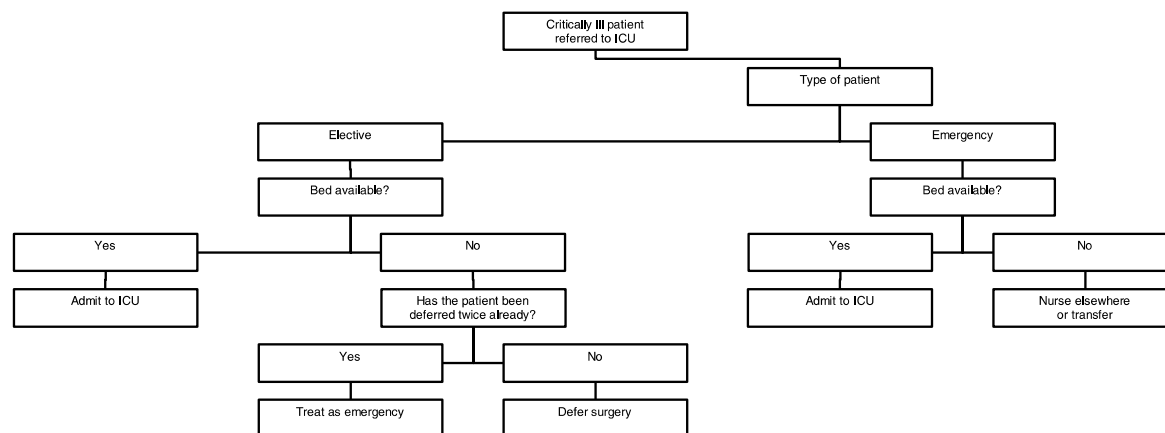


Figure 1 Flow diagram for individual patients referred to the critical care unit, used in the mathematical modelling. The decisions taken and the parameters can be changed in the model to match those for any individual critical care unit. ICU = intensive care unit.

of illness as measured by the APACHE II score, patients' age, sex and admission diagnosis might all influence their eventual length of stay. The second category of data is the arrival pattern in terms of the time and day of the week of admission to the unit. The final category is the length of stay on the unit, calculated by subtracting the date and time of admission from those of discharge.

This type of data is collected by units contributing to the Intensive Care National Audit and Research Centre's casemix programme and by many units collecting other datasets. In most critical care units, the data described above should be available.

Data analysis for the models

The initial part of the modelling involves splitting the patients according to the simple categories outlined in Fig. 1. Therefore, patients were divided into two groups: emergency or planned cases. However, the variation in length of stay with just two groups is far too large and the emergency and planned categories need dividing into further subgroups with more consistent lengths of stay. These subgroups can be formed using clinical insight, through statistical methods or through a combination of both.

Classification and Regression Tree (CART) analysis [9] is a useful statistical method for grouping patients based on a number of 'explanatory' variables with respect to a chosen criterion for the classification, such as length of stay. The ability of CART analysis to divide critically ill patients into homogeneous groups with respect to length of stay has been described previously [10]. In brief, all patients are initially put into a single group, and this is then split in two by the CART analysis on the basis of one of the explanatory variables. The resulting two groups are checked statistically to see if the split produces a significant reduction in the variation in length of stay compared with the single group. The process is then repeated with the other explanatory variables. By statistically splitting the data this way, CART analysis divides the data into subgroups as long as reductions in variation are achieved. Once further reduction in variation can no longer be achieved, the resultant group of patients is described as a terminal node. The number of patients and a fitted distribution of length of their stay are calculated for each of the terminal nodes. In this way, the whole group of critically ill patients is broken down into a series of terminal nodes containing patients with similar lengths of stay to which a mathematical distribution can be fitted.

Model structure

The basic structure of the model is essentially a complex queuing system in which patients from a variety of groups

arrive according to the calculated arrival patterns and according to the fitted statistical distributions in each of the terminal nodes. Each of the admitted patients is assigned a length of stay, which is sampled from the appropriate statistical distribution. Real arrival patterns by month, day of the week and time of day are used and the model explores all the possible combinations of admission patterns for patients. The complex nature of the problem makes it impossible to solve using mathematical equations, and so computer simulation is used with the model running repetitively to obtain a more robust estimate of what is likely to happen in real terms.

Results

Using data from a number of hospitals, the results obtained from the models were evaluated by experienced intensive care staff. We provide three examples of how the model has worked in different clinical situations. The first describes the comparison of the model's performance with actual admission patterns, whereas the second and third demonstrate the model's role in capacity planning.

Validation of the model for a critical care unit

The patient groups are shown as a 'tree of nodes', obtained from a CART analysis of 602 critically ill patients admitted to the intensive care unit of the combined Critical Care Complex at Norfolk and Norwich University NHS Trust from 01/04/00 to 31/03/01. Figure 2 shows how each node was split into two further nodes when a reduction in variation was possible. As the Critical Care Complex is a combined unit, some patients requiring only high-dependency care were nursed in the intensive care area. These patients were initially split from the remaining 486. The intensive care unit patients were then divided into emergency and planned admissions. CART used the patient's source [i.e. operating theatres, general ward, specialist areas (radiology, delivery suite) and other hospital], referring specialty and age to divide the emergency patients into six terminal nodes, each with a minimum of 50 patients. The actual and fitted distribution for one of the terminal nodes (containing the 116 high-dependency patients nursed on the intensive care area) is shown in Fig. 3.

Using the data and distributions within each terminal node, the model was run to simulate a period of one year 50 times to generate an adequate simulation. The performance of the model can be examined by comparing the model's predictions with the actual pattern of admission for the year (Table 1). The predicted and actual numbers of patients arriving and their mortality,

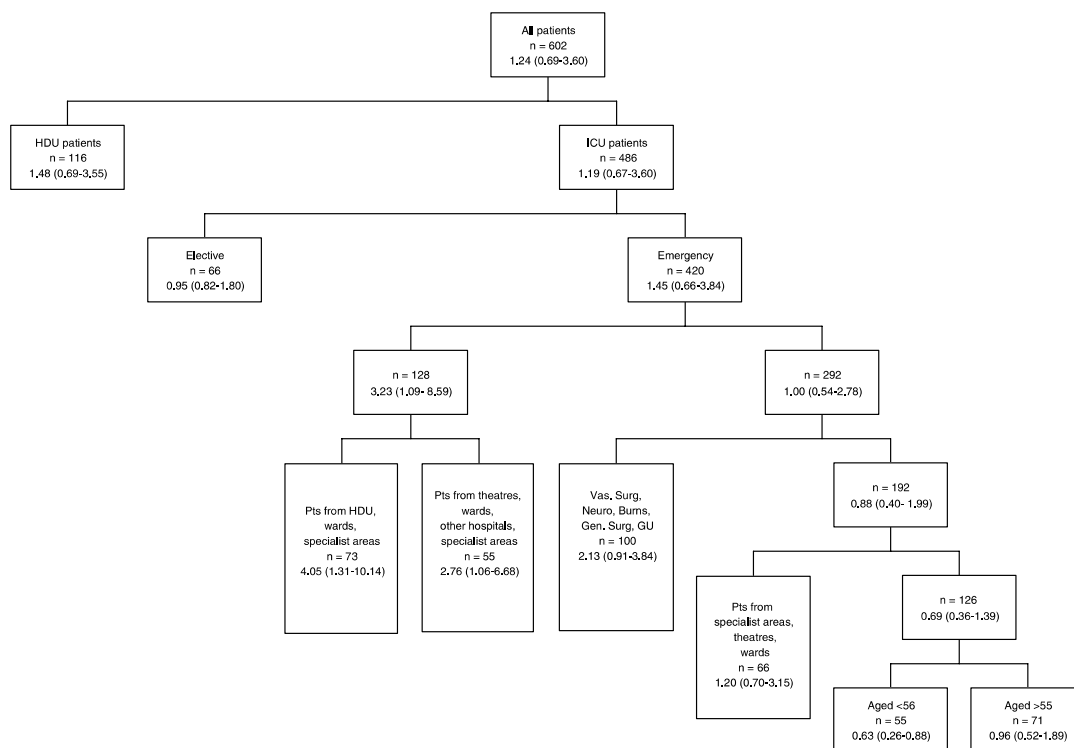
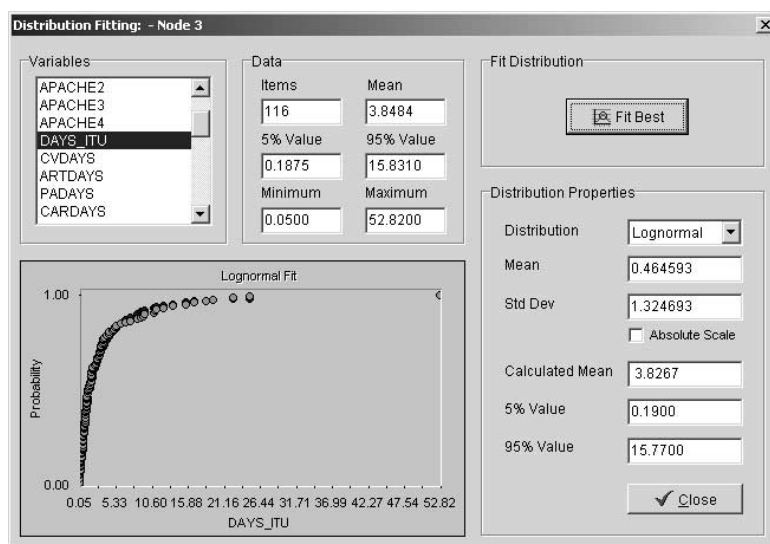


Figure 2 ‘Tree of nodes’ obtained from CART analysis of 602 patients admitted to the intensive care unit at the Norfolk and Norwich University NHS Trust. Each node shows the number of patients within the node together with their median length of stay and interquartile range (days). Note that although wards, specialist areas and theatres appear in three different terminal nodes, the patients were referred from different specialties. HDU = high dependency unit – i.e. patients requiring only high dependency care who were nursed in the intensive care area; ICU = intensive care unit; Pts = patients; Vas. Surg. = vascular surgery; Gen. Surg. = general surgery; GU = genitourinary surgery.

Figure 3 Screen capture shot of the actual and fitted distribution for the terminal node in Figure 2 (containing the 116 high dependency patients nursed on the intensive care area) when the model was run to simulate a period of one year 50 times. The best fitted distribution for this terminal node is a lognormal distribution. The concordance between the actual data and fitted distribution can be checked graphically and by comparing the means, 5% and 95% values. The distribution properties (type of distribution, mean and standard deviation) are used for generating the lengths of stay of individual patients in the simulation. Note the large variability in the lengths of stay; the recorded maximum value is 52.8 days.



bed occupancies and transfer and deferral rates were similar but the predicted number of patient-days of care was lower than the actual number. The average number

of beds in use each day is shown in Fig. 4. The distribution of the beds in use predicted by the model is narrower than that which actually occurred.

Table 1 Comparison of predicted and actual admissions to the Norfolk and Norwich University NHS Trust intensive care unit (ICU) when the model was run to simulate a period of one year 50 times. Values are number or percentage.

	Predicted	Actual
Total admissions	603	602
Emergency admissions	422	420
Elective admissions	180	182
Mortality	17.9%	17.6%
Patient-days of care; days	1940	2191
Occupancy rate:		
All eight ICU beds open (100%)	66.9%	75.0%
Seven ICU beds open (87.5%)	76.0%	85.7%
Six ICU beds open (75%)	89.2%	100.0%
Patients deferred	11.2%	2.8%*
Patients transferred	5.7%	< 1%*

*Operationally, extra ICU patients were nursed on the adjacent high dependency unit and were not recorded as deferrals or transfers.

Percentage of time in use

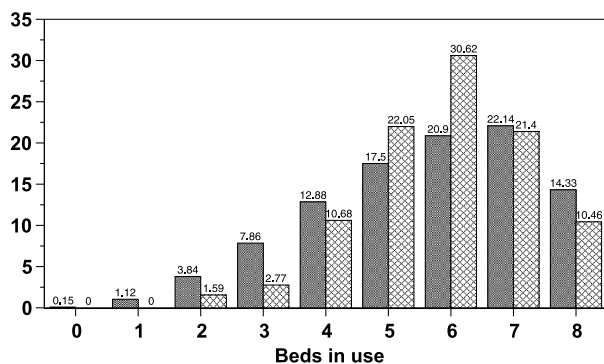


Figure 4 Predicted (solid bars) and actual (hatched bars) number of Norfolk and Norwich University NHS Trust intensive care unit beds in use each day when the model was run to simulate a period of one year 50 times.

Capacity planning

A hospital (Hospital A) provided data for intensive care use for the years 1997–2000. During this time the hospital had five intensive care beds; this was inadequate, resulting in heavy utilisation of beds and high deferral and transfer rates. The hospital wished to prepare a business case for an expansion and a decision was made to have a single critical care unit combining both intensive care and high-dependency facilities.

Following analysis of data and discussions with the senior critical care staff, four groups of intensive care patients were identified (emergency short stay, emergency normal stay, elective short stay and elective normal stay); for each of these groups, distributions of lengths of stay and arrival patterns were available. In this hospital, emergency and planned admissions had equal priorities

Table 2 Predicted results for 800 patients passing through the intensive care unit in Hospital A, using the model to support a business case for expansion. Patients are divided into 139 (17.3%) emergency short stay, 476 (59.4%) emergency normal stay, 69 (8.6%) elective short stay and 116 (14.4%) elective normal stay, with different intensive care unit capacities and two holding beds. Values are number or percentage.

	Capacity (bed numbers)			
	9	10	11	12
Total admissions	664	708	718	748
Emergency admissions	466	513	523	550
Elective admissions	198	195	195	198
Bed occupancy	84.8%	79.0%	75.3%	74.1%
Mean time in holding beds; h	9.2	8.9	8.9	8.9
Deferral rate*	36.3%	26.0%	20.0%	15.1%
Transfer rate	16.7%	11.0%	8.5%	4.9%

*Patients could be deferred more than once.

and no beds were reserved for emergency patients. However, there were two holding beds in which critically ill patients could be nursed for about 12 h while awaiting an intensive care unit bed.

Number of admissions, bed occupancy, and transfer and deferral rates predicted by the model for differently sized critical care facilities are shown in Table 2. This table shows the effects of changing bed numbers on admissions, occupancy rates, use of holding beds, deferral rate, and transfer rate when 800 new patients are expected each year. The impact of increasing bed numbers on transfers and deferrals can be clearly seen.

The second case study is for another hospital in England (Hospital B) that needed calculations to plan a combined critical care unit. The hospital database had information about intensive care and high dependency patients and the model was used to create the appropriate groups with similar lengths of stay. The model was again tuned satisfactorily in terms of type of patients to represent the particular conditions and the results of the analysis are shown in Table 3, which demonstrates that increasing capacity from 10 beds has a major impact on the transfer and deferral rates. Beyond about 12 beds, further reductions in deferral and transfer rates are only achieved at the expense of what might be considered too low an occupancy rate. Table 3 also shows the results of simple calculations for the required number of beds based on average values only. These numbers are underestimates as they imply that with the calculated bed numbers the expected number of patients will be admitted. The results from the mathematical model show that with capacities of 10–12 beds, the transfer and deferral rates would be rather high, and with 14 beds these rates would be 1.6 and 7.3%.

Table 3 Predicted results for 800 patients passing through the intensive care unit in Hospital B, requiring 3336 patient-days of care with different intensive care unit capacities. Calculations are shown using the mathematical model and using simple mathematics and Poisson distributions. Values are number or percentage.

	Capacity (bed numbers)			
	10	12	14	16
Predicted results using mathematical modelling:				
Bed occupancy	87%	76%	64%	55%
Emergency transfers	11.5%	4.6%	1.6%	0.3%
Elective deferrals	58.1%	10.3%	7.3%	1.3%
Calculated requirements using simple mathematics:				
Average no. beds required*	10.5	12.0	14.3	16.6
No. beds required to meet 95% of demand†	15.8	17.7	20.5	23.3

* $(\text{annual admissions} \times \text{average length of stay}) / (365 \times \text{occupancy})$. Average length of stay was 4.17 days in this group of patients.

† $\text{Number of beds} + 1.64 (\text{number of beds})^{1/2}$.

Table 3 also show the calculations for upper 95% CI, calculated from Poisson distributions, which are quoted sometimes in addition to the calculations based on average values.

Discussion

Current patterns of provision of critical care are the result of a largely haphazard process based on variable local factors such as funding, enthusiasm and crude estimates of need. Effective capacity planning has been largely ignored as the service developed and responded to new advances in technology and knowledge. However, because of limitations and constraints within the NHS, proper capacity planning of critical care services has become more urgent. In 2000, the Department of Health published its review of critical care services [11]; one of the review's most important recommendations was that capacity planning should take place so that crises resulting from inadequate bed availability could be minimised. Previous capacity planning for intensive care was derived from an arbitrary percentage of total hospital beds. Classically, it was recommended that critical care beds comprised up to 2% of total hospital beds but there is confusion as to whether this figure included other specialist areas within the hospital such as renal units, special care baby units or recovery areas. A simple percentage also completely ignores hospitals' casemix; those undertaking complex and specialist surgery such as vascular, cardiac or plastic surgery will require more critical care beds than hospitals that do not undertake such work. A model based only on average values is far too simple for the task of planning and managing critical care capacities.

At the outset, it was agreed that the model should be as generic as possible so that it would be applicable in most

hospitals in the UK. It is possible for individual hospitals to tune the model to represent their particular casemix, preferred organisation and care pathways. Our illustrations show how the model can be adjusted in the light of different admission criteria, priorities and expected lengths of stay. The main strength of the model, however, is its ability to use a hospital's own information about its patients so that the results are tailored to the hospital's unique casemix. Thus, the model takes detailed variability, uncertainty and complexity of a critical care service properly into account. Use of average values only for estimating the number of beds underestimates the requirements. It is not easy to interpret the statistical calculation of 95% CI meaningfully because these calculations do not really correspond to the idea that 95% of the demand will be met. Table 3 shows that these inappropriate calculations of 95% CI result in estimates that are $\approx 50\%$ higher than those predicted by the model. Most people with experience of critical care would reject these 95% values as being unrealistic estimates. The simple mathematical approach fails to capture the dynamic, yet subtle, changes in the management of critically ill patients that occur when units become full. Intensive care physicians attempt to limit further admissions, organise care elsewhere in the hospital or resort to early discharge of recovering patients. These manoeuvres will undermine a calculated bed requirement but will be accommodated in a simulation model using actual distribution of lengths of stay rather than summary statistics. Unfortunately, calculated bed requirements have been reported previously and may have inflated the perceived need for critical care beds [12].

The division into groups of patients, and the relevant statistics for the groups can be obtained very rapidly by CART analysis. The results of scenario planning can easily be tested through a series of 'what-if' manoeuvres.

Modifications in essential organisational issues, such as priorities for emergency cases and rules for admissions and transfers, can be made and the effect on bed requirements and also transfer and deferral rates explored. Provision of care through separate intensive care and high-dependency units can be tested against a combined critical care unit. The advantage of such scenario planning is that it is no longer based upon subjective estimates which may be unreliable. The independent nature of the bed estimate means that the consequences of limiting capacity can be demonstrated and discussed with those who make decisions about health care and its resources.

The weaknesses of the model mainly result from difficulty validating the results of the scenarios. Brand new critical care units are built infrequently and so opportunities for formally testing the estimates are limited. However, as the mathematical principles upon which the model is based are sound, there is no reason to believe that the predictions will not be valid, providing that the admission rules are realistic and the casemix data accurate. The distribution of simulated occupancy from the data from Norwich (Table 1 and Fig. 2) is similar to that which actually occurred; if the casemix and demand for services at Norwich does not alter in the following year, then these results would suggest that the nursing establishment should be maintained at levels which would consistently provide seven beds. Formal validation may also be undermined by the continuously evolving demand for critical care services. In the old Anglia region, the demand for critical care has increased despite increased bed capacity [13]; this suggests that with the opening of additional beds, new patient streams are identified. The model cannot compensate for new patient streams not yet identified when the model was run. However, new casemixes are easy to insert and so the model can be updated easily. Changing patterns of practice may alter the distribution of lengths of stay but once again the CART analysis is easy to repeat and significant changes in distribution easily identified. Despite these limitations, simulation modelling provides a better estimate of critical care requirements than is otherwise presently available.

Future extensions of this work could include applying the model to individual bed usage to allow workload planning in the short term. If the number of beds is known, there is no reason why the data about arrival patterns could not be applied to individual critical care beds. This may help adjustment of the daily nursing establishment and more robust booking of critical care beds for elective surgery. Detailed models can also help at regional and national levels, as models for a number of hospitals working collaboratively in a region have been developed.

Conflict of interest

Dr A.K. Shahani is also a director of Practical Insights Ltd, the company now responsible for the further development of the model. The classification program is called PORT, and the simulation program is called CCUSIM.

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Appendix

An illustration of the errors that can occur when using averages in the presence of nonlinearity and variation. The volume occupied by small and large marbles may be calculated mathematically, and such a calculation can be checked by water displacement. The use of average

diameter works well for separately estimating the total volumes of 20 small and 20 large marbles:

Small marbles

The volume of a sphere is given by $1/6\pi \times (\text{diameter})^3$. Hence the estimated total volume of 20 small marbles with an average diameter of 1.646 cm = $20 \times \{1/6\pi \times (1.646)^3\} = 47 \text{ cm}^3$.

And measured volume by water displacement = 47 cm^3 .

Large marbles

Estimated total volume of 20 large marbles with an average diameter of 2.533 cm = $20 \times \{1/6\pi \times (2.533)^3\} = 170 \text{ cm}^3$.

Measured volume by water displacement = 170 cm^3 .

However, the total volume of a mixture of 20 small and 20 large marbles is underestimated if average diameter is used:

Estimated total volume of all 40 marbles with an average diameter of 2.089 cm = $40 \times \{1/6\pi \times (2.089)^3\} = 191 \text{ cm}^3$.

But the measured volume is known to be $47 + 170 \text{ cm}^3 = 217 \text{ cm}^3$.

The variability in the diameter of 20 small and 20 large marbles is negligible, but the variability in the combined group was larger. Using only average values when there is nonlinearity and variability will result in errors; from a practical perspective, this will result in an underestimate of the true value.